

Coupling Polytopes and Toric Geometry: the 2×3 Family, the Square Case, Higher Levels and Stability

A companion to *Conditionals, Couplings and Toric Geometry*

Vinay K. Awasthi
Department of Applied Mathematics and Statistics
Johns Hopkins University*

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Abstract

A coupling of two finite distributions is a point of a transportation polytope. For nondegenerate marginals this polytope is Delzant [2], so it is the moment polytope of a compact toric symplectic manifold. This companion works the correspondence out in the cases that can be computed completely, and records how far it goes.

- *The 2×3 family.* The six walls $f_i = g_j$ cut the space of marginals into 18 chambers, which fall into 5 orbits under the symmetry group $S_2 \times S_3$. The surfaces that occur are $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, the Hirzebruch surface F_1 , $\mathbb{CP}^2$ blown up at two points, and the del Pezzo surface of degree 6. Every facet normal lies in the fan of the last, so every surface in the family is the del Pezzo surface or a toric blow-down of it. Crossing a single wall is exactly one toric blow-up or blow-down.
- *The Kähler class.* It is the product coupling read as divisor coefficients, $[\omega_{f,g}] = \sum_c f_i g_j [D_c]$. Edge lengths of the coupling polygon are degrees of ω on toric curves, and its area is the symplectic volume.
- *The square 2×2 case.* With the unital whiskering, explicit whiskering defects are nonzero, and every defect is a tangent vector of a coupling polytope.
- *Higher levels.* The Delzant theorem applies at levels two and three. Flattening a level-two coupling is affine and commutes with stochastic change of base, but it is neither surjective nor injective in general.
- *Disc potentials.* They are sums over facets, which are cells, not cycles. For the degree-6 del Pezzo surface the potential has six critical points, all over the product coupling, with critical values $6, -2, -2, -2, -3, -3$. Square nondegenerate polytopes with $n \geq 3$ give manifolds that are not Fano. Optimal transport minimizes a moment-map component, so its optima are torus-fixed points, that is, spanning-tree couplings.
- *Stability.* Each $\omega_{f,g}$ gives a Bridgeland stability condition on the derived category of the corresponding surface, with explicit central charges. We state what this does and does not connect.
- *Cumulants.* Centred couplings are second mixed cumulants, and level-two flattening keeps only the cross-covariance of random measures. The whiskering defects are the Boolean cumulants of whiskering, viewed as a noncommutative expectation. They are circulations, so tangent vectors of coupling polytopes. A cone in the sense of homotopy probability theory makes whiskering an A_∞ -functor through arity three, and its cohomology keeps only the marginals.

*Verification script: `check_polytope_companion.py` (26 checks; exact rational arithmetic except where marked). This version replaces the draft “Coupling polytopes and the 2×3 family”; [appendix A](#) lists what was corrected and what was withdrawn.

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1 Introduction

The companion paper [2] proves that the polytope $\text{Cpl}(f, g)$ of couplings of two nondegenerate marginals is Delzant. By Delzant’s theorem [8] there is therefore a compact toric symplectic manifold $X_{f, g}$ whose moment map has image $\text{Cpl}(f, g)$. Each interior coupling is the image of a Lagrangian torus, and each vertex is the image of a fixed point. This note asks what the correspondence gives in concrete cases.

Section 2 fixes conventions and proves a general formula for the Kähler class. Section 3 treats the 2×3 family completely. Section 4 treats the square 2×2 case, the base case for the whiskering questions of [2, §7]. Section 5 extends the polytope results to second- and third-order states. Section 6 computes disc potentials and relates optimal transport to the moment map. Section 7 constructs stability conditions from marginals and states the remaining questions. Section 8 records the status of the A_∞ question, and section 9 connects the polytopes to cumulants and homotopy probability theory. Every numbered claim is checked in `check_polytope_companion.py`.

2 Setup

Let $f \in D(Y)$ and $g \in D(Y')$ have full support, with $|Y| = n$ and $|Y'| = m$, and let

$$\text{Cpl}(f, g) = \left\{ \alpha \in \mathbb{R}_{\geq 0}^{Y \times Y'} : \sum_{y'} \alpha(y, y') = f(y), \sum_y \alpha(y, y') = g(y') \right\}.$$

The tangent space is the space of circulations, that is matrices with zero row and column sums. The integer circulations form a lattice Λ of rank $(n-1)(m-1)$ with basis $e_{ij} = E_{ij} - E_{im} - E_{nj} + E_{nm}$ ($i < n, j < m$). Coordinates in this basis are the entries in the top-left $(n-1) \times (m-1)$ block. The pair (f, g) is *degenerate* if $\sum_I f = \sum_J g$ for some nonempty proper $I \subsetneq Y$ and $J \subsetneq Y'$. These hyperplanes are the *walls*, and the components of their complement are the *chambers*.

Theorem 2.1 ([2, Thm. 4.4]). *If (f, g) is nondegenerate, every vertex of $\text{Cpl}(f, g)$ is supported on a spanning tree of $K_{n,m}$, the edges at a vertex are the fundamental cycles of that tree, and $\text{Cpl}(f, g)$ is Delzant with respect to Λ .*

Facets and normals. The facets are among the sets $\{\alpha_c = 0\}$, one for each cell $c = (i, j)$. A cell gives no facet when $f_i + g_j > 1$, since then $\alpha_c \geq f_i + g_j - 1 > 0$. The inward normal of $\{\alpha_c = 0\}$ is the coordinate functional $\alpha \mapsto \alpha_c$ restricted to Λ . In the dual basis $\{e_{ij}^*\}$ it is

$$v_{(i,j)} = e_{ij}^*, \quad v_{(i,m)} = -\sum_j e_{ij}^*, \quad v_{(n,j)} = -\sum_i e_{ij}^*, \quad v_{(n,m)} = \sum_{i,j} e_{ij}^* \quad (i < n, j < m).$$

Each v_c is primitive, because some integer circulation has entry 1 at c .

Proposition 2.2 (The Kähler class is the product coupling). *For nondegenerate (f, g) , let D_c be the toric divisor of the facet $\{\alpha_c = 0\}$. Up to the normalization of ω , the symplectic class of $X_{f,g}$ is*

$$[\omega_{f,g}] = \sum_{c \text{ facet}} f_i g_j [D_c], \quad c = (i, j),$$

and $\omega_{f,g} \cdot C$ is the lattice length of the edge of $\text{Cpl}(f, g)$ that corresponds to a torus-invariant curve C .

Proof. For a Delzant polytope $\{u : \langle u, v_c \rangle + \lambda_c \geq 0\}$ the symplectic class is $\sum_c \lambda_c [D_c]$ [12]. Here the facet functions are $\alpha_c(u) = \langle u - u_0, v_c \rangle + \alpha_c(u_0)$ for any interior u_0 , so $\lambda_c = \alpha_c(u_0)$. Taking $u_0 = f \otimes g$ gives the formula. Another base point changes the coefficients by $\langle w, v_c \rangle$, which is a linear relation among the $[D_c]$. The degree of ω on a torus-invariant curve is the lattice length of its edge. $\square$

3 The 2×3 family

Write $f = (f_1, 1 - f_1)$, $g = (g_1, g_2, 1 - g_1 - g_2)$, $x = \alpha_{11}$ and $y = \alpha_{12}$. The polygon $\text{Cpl}(f, g)$ is cut out by

$$x \geq 0, \quad y \geq 0, \quad x \leq g_1, \quad y \leq g_2, \quad x + y \leq f_1, \quad x + y \geq f_1 + g_1 + g_2 - 1,$$

with inward normals $e_1, e_2, -e_1, -e_2, -(e_1 + e_2)$ and $e_1 + e_2$. The only proper subsets of a two-point and a three-point set that can have equal sums give the six walls $f_i = g_j$.

Theorem 3.1 (Chambers and surfaces). *The six walls cut the prism $\{0 < f_1 < 1, g_1, g_2 > 0, g_1 + g_2 < 1\}$ into 18 chambers. On each chamber the polygon type is constant and Delzant, and the surfaces are:*

Polygon	Chambers	Self-intersections D_c^2	Surface	Orbit under $S_2 \times S_3$
triangle	2	(1, 1, 1)	$\mathbb{CP}^2$	one orbit of size 2
quadrilateral	3	(0, 0, 0, 0)	$\mathbb{CP}^1 \times \mathbb{CP}^1$	one orbit of size 3
quadrilateral	6	(1, 0, -1, 0)	$F_1 = \text{Bl}_1 \mathbb{CP}^2$	one orbit of size 6
pentagon	6	(0, -1, -1, -1, 0)	$\text{Bl}_2 \mathbb{CP}^2$	one orbit of size 6
hexagon	1	$(-1)^6$	del Pezzo of degree 6	fixed

Here S_2 swaps the two rows and S_3 permutes the columns.

Proof. For an arrangement of hyperplanes restricted to a convex region, each realizable sign vector is exactly one convex chamber. A rational grid and random sampling realize the same 18 sign vectors. In each chamber the vertices are computed exactly, and the number of vertices and the self-intersections are constant on all samples. Self-intersections come from the relations $v_{i-1} + v_{i+1} = -D_i^2 v_i$ between cyclically ordered normals. Consistency is confirmed by $\sum_i D_i^2 = 12 - 3k$ for k divisors. Delzant means that consecutive normals have determinant ± 1 . The orbits are computed by letting $S_2 \times S_3$ act on sign vectors. $\square$

Theorem 3.2 (Every surface is a blow-down of the del Pezzo surface). *For every nondegenerate pair in the 2×3 family, $X_{f,g}$ is the del Pezzo surface of degree 6 or is obtained from it by toric blow-downs.*

Proof. The fan Σ of $X_{f,g}$ is smooth and complete. Its rays are among the six rays $\pm e_1, \pm e_2, \pm(e_1 + e_2)$ of the del Pezzo fan Σ_6 . In cyclic order each ray of Σ_6 is the sum of its neighbours. Two of the six rays form a lattice basis exactly when they are at cyclic distance one or two; rays at distance three are opposite. So adjacent rays u, w of Σ are at distance one or two, and in the second case the omitted ray between them is $u + w$. Inserting $u + w$ is the star subdivision that blows up the corresponding fixed point [10]. Inserting all omitted rays turns Σ into Σ_6 . $\square$

Proposition 3.3 (Wall-crossing). *On the wall $f_i = g_j$ three facet lines meet at a point. For example, on $f_1 = g_1$ the lines $x = g_1, y = 0$ and $x + y = f_1$ meet at $(g_1, 0)$. Crossing one wall cuts or restores one corner, which is one toric blow-up or blow-down. There are no flips, which do not occur for surfaces, and the type does not change inside a chamber.*

Proof. The concurrency is direct. For all 30 pairs of chambers whose sign vectors differ in one wall, the vertex counts differ by exactly one. Constancy inside a chamber is [2, Prop. 4.6]. $\square$

Example 3.4.

- (a) At $(f_1, g_1, g_2) = (0.1, 0.5, 0.2)$ the polygon is the triangle with vertices $(0, 0)$, $(0.1, 0)$ and $(0, 0.1)$, and the surface is $\mathbb{CP}^2$.
- (b) At $(0.4, 0.5, 0.3)$ the polygon is the pentagon with vertices $(0, 0.2)$, $(0, 0.3)$, $(0.1, 0.3)$, $(0.4, 0)$ and $(0.2, 0)$, and the surface is $\text{Bl}_2 \mathbb{CP}^2$. The straight path to the uniform point crosses only the wall $f_1 = g_1$; the corner at $(f_1, 0)$ is cut and the hexagon appears.
- (c) At the uniform point $(\frac{1}{2}, \frac{1}{3}, \frac{1}{3})$ the hexagon has vertices $(0, \frac{1}{6})$, $(0, \frac{1}{3})$, $(\frac{1}{6}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{6})$, $(\frac{1}{3}, 0)$ and $(\frac{1}{6}, 0)$, all six edges have lattice length $\frac{1}{6}$, its lattice area is $\frac{1}{12}$, and $[\omega] = -\frac{1}{6}K_X$. This is the unique monotone point of the family [2, Prop. 4.7]: monotonicity holds at a point, not on a chamber.

Proposition 3.5 (Kähler data in the family). *In all 18 chambers, $\omega_{f,g} \cdot D_c$ equals the lattice length of the edge $\{\alpha_c = 0\}$ and is positive, so $\omega_{f,g}$ is ample, and $\frac{1}{2}\omega_{f,g}^2$ equals the lattice area of $\text{Cpl}(f, g)$.*

Proof. These are proposition 2.2 and the Duistermaat–Heckman formula for the volume. The script verifies both in exact arithmetic, using the intersection matrix of each fan. $\square$

4 The square 2×2 case

Let $B = B' = \{0, 1\}$. A 1-cell is a stochastic matrix $k(a, b)$ with rows $(1 - a, a)$ and $(1 - b, b)$, and Kleisli composition is matrix multiplication. The deterministic 1-cells are the constants $e_0 = k(0, 0)$ and $e_1 = k(1, 1)$, the identity $\sigma = k(0, 1)$ and the swap $\tau = k(1, 0)$. A coupling 2-cell $\alpha: f \Rightarrow g$ is a coupling $\alpha(x)$ of $f(x)$ and $g(x)$ at each x . Its cost is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$. Vertical composition is the gluing lemma, strictly associative with diagonal identities [2, Thm. 2.3].

Proposition 4.1 (Coupling segments and costs). *For $f(x) = (p, 1 - p)$ and $g(x) = (q, 1 - q)$, the couplings are $\begin{pmatrix} a & p-a \\ q-a & 1-p-q+a \end{pmatrix}$ with $\max(0, p + q - 1) \leq a \leq \min(p, q)$. The product coupling has $a = pq$, and the maximal coupling, of cost $\text{TV}(f(x), g(x))$, has $a = \min(p, q)$. Between deterministic 1-cells the coupling is unique, and its cost is 0 for equal 1-cells and 1 for any two distinct ones, because distinct deterministic maps differ at some x and the cost is a maximum over x .*

Definition 4.2 (Whiskering). Precomposition by a deterministic $u: W \rightarrow B$ is $(\alpha \circ u)(w) = \alpha(u(w))$. Postcomposition by $h: B' \rightarrow D(B'')$ copies h on equal pairs and runs it independently on unequal pairs:

$$(h \cdot \alpha)(x)(z, z') = \sum_y \alpha(x)(y, y) h(y)(z) [z = z'] + \sum_{y \neq y'} \alpha(x)(y, y') h(y)(z) h(y')(z').$$

This is the whiskering of [2, Thm. 2.3(ii)]. It sends diagonal couplings to diagonal couplings. Running h independently also on equal pairs would not preserve identities: for $h = k(\frac{9}{10}, \frac{1}{10})$ it sends $\frac{1}{2}\Delta$ to $(\frac{0.41}{0.09}, \frac{0.09}{0.41})$, so it cannot be the first component of any functor.

Proposition 4.3 (Whiskering defects). *Let $F_2(\beta, \alpha) = (h \cdot \beta) \cdot (h \cdot \alpha) - h \cdot (\beta \cdot \alpha)$.*

- (i) *Precomposition by a deterministic map is strictly functorial.*
- (ii) *If α is an identity, then $F_2(\beta, \alpha) = 0$.*
- (iii) *With $f = g$ having uniform rows and $\alpha = \beta$ the swap coupling at each x , $F_2(\beta, \alpha)(x) = t \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$, where $t = \frac{369}{2500}$ for $h = k(\frac{9}{10}, \frac{1}{10})$ and $t = \frac{1}{4}$ for h constant and uniform.*
- (iv) *Every defect has zero row and column sums. It is a tangent vector of the coupling polytope of $h^\# f(x)$ and $h^\# k(x)$, not a coupling.*

Proof. (i) Both sides evaluate the same couplings at $u(w)$. (ii) $h \cdot \Delta = \Delta$, and Δ is a unit for gluing. (iii) $\beta \cdot \alpha = \text{swap} \cdot \text{swap} = \Delta$, so $h \cdot (\beta \cdot \alpha)$ is diagonal, while $h \cdot \text{swap}$ has mass off the diagonal; the values are exact computations. (iv) Both terms couple the same two marginals. $\square$

So the defect lives in the tangent space of a coupling polytope. There the Fisher metric, which is twice the Hessian of Guillemin's potential [2, Prop. 4.5], gives it a size. This is the only sense in which the defect meets the toric geometry that we can establish.

5 Second- and third-order states

Let $Y \subseteq DB$ be a finite set of n first-order states and $P, Q \in D(Y)$ second-order states with full support. A level-two coupling is $\Pi \in D(Y \times Y)$ with marginals P and Q , and $\text{Cpl}_2(P, Q)$ is the transportation polytope on $K_{n,n}$ with these marginals.

Proposition 5.1 (Delzant at levels two and three). *If (P, Q) is nondegenerate, then $\text{Cpl}_2(P, Q)$ is Delzant of dimension $(n - 1)^2$. The same holds for third-order states $R, S \in \mathcal{D}(Z)$, with Z a finite set of second-order states.*

Proof. [Theorem 2.1](#) uses only the marginals, not what the points of Y or Z are. $\square$

Flattening. The map

$$\mu^*: \text{Cpl}_2(P, Q) \rightarrow \text{Cpl}(\mu_B P, \mu_B Q), \quad \mu^*(\Pi)(b, b') = \sum_{\rho, \rho'} \Pi(\rho, \rho') \rho(b) \rho'(b'),$$

is affine. Its image contains $\mu_B P \otimes \mu_B Q$, the image of $P \otimes Q$.

Proposition 5.2 (Properties of flattening).

- (i) *If Y consists of point masses, μ^* is the identity.*
- (ii) *μ^* is not surjective in general: for $Y = \{\rho\}$ with ρ uniform on two points the image is $\{\rho \otimes \rho\}$, while $\text{Cpl}(\rho, \rho)$ is a segment containing the diagonal coupling.*
- (iii) *For $n = 2$ and $\rho_1 \neq \rho_2$, μ^* is injective: the segment direction maps to $(\rho_1 - \rho_2) \otimes (\rho_1 - \rho_2) \neq 0$.*
- (iv) *If $(n - 1)^2 > (|B| - 1)^2$, the fibres of μ^* are positive-dimensional. For example, with three states on two outcomes and uniform $P = Q$ the fibre has dimension 3, so level-two couplings carry information that level-one couplings lose.*
- (v) *For a kernel $k: B \rightarrow \mathcal{D}(B')$, pushing Π forward along $k^\# \times k^\#$ commutes with flattening: $\mu^*((k^\# \times k^\#)_* \Pi) = (k \otimes k)^\# \mu^*(\Pi)$. This is the level-two form of stochastic change of base [4, Prop. 9.14].*

Proof. (i) $\rho(b) = [\rho = \delta_b]$. (ii) and (iii) are direct. (iv) The image lies in a space of dimension $(|B| - 1)^2$; the example is computed exactly. (v) Both sides equal $\sum \Pi(\rho, \rho') k^\# \rho \otimes k^\# \rho'$. $\square$

Uniform marginals. Uniform $n \times n$ marginals are degenerate for every $n \geq 2$. For $n = 2$ the polytope is still a segment, so $X = \mathbb{CP}^1$ and it is monotone, with special point the product coupling. For $n \geq 3$ the polytope is not simple and has no smooth toric manifold.

6 Disc potentials and optimal transport

For a Delzant polytope with facet functions ℓ_c and normals v_c , the potential of the torus fibre over u is $W_u(z) = \sum_c T^{\ell_c(u)} z^{v_c}$ over facets, for compact toric Fano manifolds [7] and, with corrections, for general compact toric manifolds [9]. A fibre with a flat connection has nonvanishing Floer cohomology when it corresponds to a critical point. For coupling polytopes the facets are cells, so W has at most nm terms, with exponents given by the normals of [section 2](#). The facets are not the cycles of $K_{n,m}$.

Proposition 6.1 (The del Pezzo potential). *At the uniform 2×3 point all facet functions equal $\frac{1}{6}$ at the product coupling, and $W = T^{1/6}(x + y + x^{-1} + y^{-1} + xy + x^{-1}y^{-1})$. Up to the factor $T^{1/6}$, W has six critical points: $(1, 1)$, $(1, -1)$, $(-1, 1)$, $(-1, -1)$, (ζ, ζ) and (ζ^2, ζ^2) for a primitive cube root of unity ζ . All six have $|x| = |y| = 1$, so they lie over the product coupling and differ only in the flat connection. The critical values are $6, -2, -2, -2, -3, -3$.*

Proof. Exact solution of $x\partial_x W = y\partial_y W = 0$; for instance $W(\zeta, \zeta) = 3\zeta + 3\zeta^2 = -3$. The number 6 equals twice the area 3 of the Newton hexagon, as Kushnirenko's theorem predicts. $\square$

Example 6.2 (Non-monotone cases).

- (a) *A segment.* For the 2×2 segment $[\max(0, p+q-1), \min(p, q)]$, $X = \mathbb{CP}^1$ and the special fibre is the midpoint. That is the product coupling $a = pq$ when $p = \frac{1}{2}$, but not in general: for $p = q = \frac{1}{3}$ the midpoint is $\frac{1}{6}$ and $pq = \frac{1}{9}$.
- (b) *A 3×3 case.* For $P = (0.40, 0.35, 0.25)$ and $Q = (0.30, 0.43, 0.27)$ the polytope is Delzant with 18 vertices and 9 facets, one per cell, so W has 9 terms.

Proposition 6.3 (Square cases are not Fano). *Let $n \geq 3$ and let (f, g) be a nondegenerate $n \times n$ pair for which every cell is a facet. Then $X_{f,g}$ is not Fano. In the 3×3 example above, $-K_X$ is nef and big, so X is weak Fano.*

Proof. $-K_X = \sum_c D_c$ corresponds to the polytope $\{\langle u, v_c \rangle \geq -1 \ \forall c\}$, which is the transportation polytope with all row and column sums n , that is n times the Birkhoff polytope. If $-K_X$ were ample, this polytope would have normal fan equal to the smooth fan of X and would be simple, but the Birkhoff polytope is not simple for $n \geq 3$. For nefness, each maximal cone (a spanning tree T) must give a vertex of that polytope, that is, the tree solution of T with all marginals equal to n must be nonnegative. This is checked for all 18 trees of the example. Bigness is full-dimensionality. $\square$

So Cho–Oh's Fano theorem applies to the 2×3 family, whose surfaces are all Fano, but not to square cases with $n \geq 3$, where the general theory of [9] is needed.

Proposition 6.4 (Optimal transport is a moment-map problem). *For a cost $c: Y \times Y' \rightarrow \mathbb{R}$, the optimal-transport value $\min_{\alpha \in \text{Cpl}(f,g)} \langle c, \alpha \rangle$ is the minimum over $X_{f,g}$ of the function $\langle c, \mu \rangle$, where μ is the moment map. It is attained at a torus-fixed point, that is, a vertex: a coupling supported on a spanning tree. If c is integral on Λ , then $\langle c, \mu \rangle$ is the Hamiltonian of a circle action.*

Proof. μ maps $X_{f,g}$ onto $\text{Cpl}(f, g)$. A linear function on a polytope is minimized at a vertex, and the preimage of a vertex is a fixed point. A component $\langle \xi, \mu \rangle$ with ξ in the lattice generates the circle action of ξ [8]. The script checks the first statement by linear programming against vertex enumeration. $\square$

This is where the polytope picture meets the obstruction theory of [4]. Kantorovich comparisons between second-order states [4, Prop. 9.14], and distances to effective sets, are linear programs over polytopes of this kind. Their optima are spanning-tree couplings, and their dual variables are the certificates. These vertices are not “committed” states in the tower's sense. A spanning-tree coupling has $n + m - 1$ nonzero entries and is not a point mass.

7 Stability conditions

A Bridgeland stability condition on a triangulated category is a pair $(Z, \mathcal{A})$: the heart $\mathcal{A}$ of a bounded t -structure, and an additive central charge $Z: K_0 \rightarrow \mathbb{C}$ that maps nonzero objects of $\mathcal{A}$ into the upper half-plane together with $\mathbb{R}_{<0}$, with Harder–Narasimhan filtrations and the support property [5]. The stability conditions form a complex manifold [5]. On any smooth projective surface, for any ample class ω , there is one with central charge $Z_\omega(E) = -\int e^{-i\omega} \text{ch}(E)$ and a tilted heart [1, 16]. (Bridgeland's earlier construction [6] is for K3 surfaces.)

Proposition 7.1 (Marginals give stability conditions). *For (f, g) in a chamber of the 2×3 family, $\omega_{f,g}$ defines a stability condition $\sigma_{f,g}$ on $D^b(\text{Coh } X_{f,g})$, depending continuously on (f, g) within the chamber, with*

$$Z(\mathcal{O}_X) = \text{area Cpl}(f, g), \quad Z(\mathcal{O}_x) = -1, \quad Z(\mathcal{O}_{D_c}) = \frac{1}{2}D_c^2 + i \text{ length}\{\alpha_c = 0\}.$$

As (f, g) approaches a wall from inside a chamber, one edge length tends to 0, its divisor E is a (-1) -curve, and $Z(\mathcal{O}_E) \rightarrow -\frac{1}{2}$.

Proof. $\omega_{f,g}$ is ample by [proposition 3.5](#). The values follow from $\text{ch}(\mathcal{O}_X) = 1$, $\text{ch}(\mathcal{O}_x) = [\text{pt}]$ and $\text{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$, together with [proposition 3.5](#). The statement about walls is [proposition 3.3](#) with the self-intersections of [theorem 3.1](#), and is checked in [2, §7.1]. $\square$

Mirror symmetry. For a toric Fano manifold the disc potential is the Hori–Vafa Landau–Ginzburg potential [7]. The disc potential enters central charges through oscillatory integrals $\int e^{-W/z}$ over integral cycles, whose asymptotics involve the $\widehat{\Gamma}$ -class [15, 11], not through its values. The critical values in [proposition 6.1](#) are the eigenvalues of quantum multiplication by c_1 , which is what the Gamma conjectures concern.

Question 7.2.

- (i) Does the path $\sigma_{f,g}$ continue across a wall, glued along Orlov’s decomposition $D^b(\text{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X) \rangle$ [17]?
- (ii) Is there a natural object attached to an interior coupling, for instance the mirror of its torus fibre with a flat connection, whose central charge has a probabilistic meaning?
- (iii) Do the quantum-corrected central charges with $\omega = \omega_{f,g}$ define stability conditions on these surfaces, and does the monotone point play a special role?

Not claimed. We make no identification of the disc potential with a central charge, of spanning-tree couplings with a heart, of the obstruction tower with Harder–Narasimhan filtrations, or of hallucination with instability.

8 The A_∞ question

The hom-categories of coupling 2-cells are strictly associative, so as A_∞ -categories they have $m_k = 0$ for $k \geq 3$. If a linearization also has $m_1 = 0$, the arity-two A_∞ -functor equation reduces to $F_1(m_2(\beta, \alpha)) = m_2(F_1\beta, F_1\alpha)$, which says that whiskering is functorial. [Proposition 4.3\(iii\)](#) shows that it is not. Hence the defect can be the second component of an A_∞ -functor only in a linearization with nonzero m_1 , for instance chains on coupling polytopes in which the defects are boundaries [2, Conj. 7.3]. No such linearization is known. Statements identifying the defect with the Hessian of a disc potential, or the functor equation with a Picard–Lefschetz formula, are withdrawn ([appendix A](#)). [Section 9](#) shows how a cone supplies such an m_1 .

9 Cumulants and the homotopy viewpoint

Homotopy probability theory treats an expectation that fails to be an algebra map as a morphism of homotopy algebras. Its components are then the cumulants [13, 14]. This section shows that the

coupling polytopes carry exactly this structure. Couplings are second mixed cumulants, level-two flattening keeps only second cumulants, and the whiskering defects are the Boolean cumulants of the whiskering map. All of them are tangent vectors of coupling polytopes. Everything here is checked in `check.cumulants.py`.

Proposition 9.1 (Couplings are second cumulants). *For $\alpha \in \text{Cpl}(f, g)$, the centred coupling $\alpha - f \otimes g$ is the cross-covariance matrix $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$ of the indicator vectors under α . So $\text{Cpl}(f, g) - f \otimes g$ is the set of cross-covariances compatible with the marginals and with positivity. It lies in the space of circulations, and the product coupling is the point where every mixed cumulant of the pair (y, y') vanishes. The product coupling is also the only point of $\text{Cpl}(f, g)$ of rank one, that is, the only point on the toric variety of the independence model. This is the Birch point of that model [18].*

Proof. $\mathbb{E}[1_{\{y=i\}}1_{\{y'=j\}}] - \mathbb{E}[1_{\{y=i\}}]\mathbb{E}[1_{\{y'=j\}}] = \alpha_{ij} - f_i g_j$. A nonnegative rank-one matrix $uv^\top$ with row sums f and column sums g equals $f \otimes g$. For two finite variables, independence is equivalent to vanishing of all mixed cumulants of the indicators. $\square$

Proposition 9.2 (Level two: flattening keeps the second cumulant). *For a level-two coupling $\Pi \in \text{Cpl}_2(P, Q)$,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \text{Cov}_\Pi(\rho, \rho'),$$

where $\text{Cov}_\Pi(\rho, \rho') = \mathbb{E}_\Pi[(\rho - \mu_B P) \otimes (\rho' - \mu_B Q)]$ is the cross-covariance of the two random measures, a circulation. The fibres of μ^ are therefore the level-two couplings with a fixed cross-covariance. For three states on two outcomes, the mixed moments $\mathbb{E}_\Pi[\rho(0)^a \rho'(0)^b]$ with $a, b \leq 2$ determine Π , while flattening sees only $a = b = 1$; this accounts for the three-dimensional fibres of [proposition 5.2\(iv\)](#).*

Proof. Expand $\mathbb{E}_\Pi[\rho \otimes \rho']$. For the example, the map from the four-dimensional space of circulations to the mixed moments of order up to $(2, 2)$ has rank 4, and to the order- $(1, 1)$ moment it has rank 1. $\square$

So level-two couplings carry the higher mixed cumulants of a pair of random distributions, and level-one gluing keeps only the second. This is the cumulant form of “level-two gluing keeps correlations that level-one gluing loses” [3, Prop. 7.16].

Whiskering as an expectation. Fix $f \in D(Y)$ and let A_f be the space of real $Y \times Y$ matrices whose row sums and column sums are both multiples of f ; it is spanned by the couplings of f with itself. With the gluing product $a \cdot b = a \text{diag}(f)^{-1} b$ and unit $\text{diag}(f)$, A_f is an associative unital algebra. The circulations I_f form a two-sided ideal, and $A_f/I_f \cong \mathbb{R}$ records only the total mass. Whiskering by a kernel h ([definition 4.2](#)) is a linear unital map $h_*: A_f \rightarrow A_{h\#f}$, a noncommutative “expectation” that need not be multiplicative. Its *Boolean cumulants* [19] are defined recursively over interval partitions:

$$\begin{aligned} b_1 &= h_*, & b_2(a, b) &= h_*(a \cdot b) - h_*(a) \cdot h_*(b), \\ b_3(a, b, c) &= h_*(a \cdot b \cdot c) - h_*a \cdot h_*b \cdot h_*c - b_2(a, b) \cdot h_*c - h_*a \cdot b_2(b, c). \end{aligned}$$

Here $a \cdot b$ means: glue a first, then b .

Proposition 9.3 (Whiskering defects are Boolean cumulants).

(i) *The whiskering defect of [proposition 4.3](#) is $F_2(\beta, \alpha) = -b_2(\alpha, \beta)$.*

- (ii) Every b_n with $n \geq 2$ takes values in the circulations $I_{h\#f}$, that is, in tangent vectors of coupling polytopes. Modulo $I_{h\#f}$, h_* is an algebra map.
- (iii) For $|Y| = 3$ the defects b_2 span all of $I_{h\#f}$, and b_3 is nonzero in general.
- (iv) Let C be the differential graded algebra obtained from $A_{h\#f}$ by freely adjoining generators $\delta(a, b)$ of degree -1 with $d\delta(a, b) = b_2(a, b)$; this is a cone in the sense of homotopy probability theory. Then (h_*, δ) satisfies the arity-two equation of an A_∞ -functor $A_f \rightarrow C$, the arity-three obstruction is a cycle, and $H^0(C) = A_{h\#f}/(b_2)$, which equals $A_{h\#f}/I_{h\#f} \cong \mathbb{R}$ whenever the defects span $I_{h\#f}$, as in (iii).

Proof. (i) is the definition. (ii) Both composites couple the same two marginals, so their difference is a circulation; the recursion preserves this because I is an ideal. Modulo I only masses remain, and they multiply. (iii) The defects from 40 random instances have rank $4 = \dim I$ for three points, and random instances give $b_3 \neq 0$. (iv) The arity-two equation is $d\delta = b_2$. The arity-three obstruction is $h_*a \cdot \delta(b, c) - \delta(a, b) \cdot h_*c - \delta(a \cdot b, c) + \delta(a, b \cdot c)$, whose differential vanishes identically; this is verified exactly. H^0 of a free extension by degree -1 generators is the quotient by the two-sided ideal generated by their boundaries, here $I_{h\#f}$ by (iii). $\square$

What this shows. Making whiskering an A_∞ -functor through a cone forgets exactly the correlations. The cohomology H^0 keeps only the marginals, that is, the Kleisli level of [2]. The correlation structure lives in the homotopy data: the Boolean cumulants b_n , which are tangent vectors of the coupling polytopes. In the geometry of section 2, the base of the fibration $\text{Cpl}(f, g) \mapsto (f, g)$ is the cohomological part, and the fibres, where the Fisher metric measures the cumulants [2, Prop. 4.5], carry the homotopy. This also explains section 8: with $m_1 = 0$ the cumulants have nowhere to live, and a cone supplies the missing m_1 .

Question 9.4.

- (i) Do the higher Boolean cumulants b_n , with generators of degree $1 - n$, complete (h_*, δ) to an A_∞ -functor into a cone, as in the bar-construction description of Boolean cumulants? This is checked here only through arity three.
- (ii) Classical and free cumulants are organized by set partitions and non-crossing partitions, and A_∞ -structures by associahedra. The permutohedron is a linear image of the Birkhoff polytope, and its faces correspond to ordered set partitions. Uniform square coupling polytopes are scaled Birkhoff polytopes, which lie on the walls of the chamber structure. Does cumulant combinatorics live on these degenerate walls?

A Changes from the earlier draft

Earlier claim	Status
22 chambers; 4/12/6 quadrilaterals, pentagons, hexagons	18 chambers; 2 triangles, 9 quadrilaterals, 6 pentagons, 1 hexagon (theorem 3.1). The accompanying script printed 18; a coordinate-order bug in its <code>intersect</code> produced wrong vertex counts.
Example 4.2 a quadrilateral	a triangle, $\mathbb{CP}^2$ (example 3.4)
Self-intersection table	corrected; the earlier patterns violated $\sum D_i^2 = 12 - 3k$
Flips; type change across $f_1 = \frac{1}{2}$	neither occurs (proposition 3.3)
Costs $c(\sigma, e_0) = c(e_0, \tau) = \frac{1}{2}$	both are 1; cost is a maximum over x
Postcomposition running h independently on equal pairs	not unital; replaced by definition 4.2 . The earlier example then has zero defect, and new examples are nonzero (proposition 4.3)
Flattening surjective; Example 10.10 constant	not surjective; for point masses it is the identity (proposition 5.2)
Uniform level two monotone iff every cell is a facet; $n = 2$ not monotone	uniform $n \times n$ is degenerate; $n = 2$ is monotone, $n \geq 3$ is not simple
Level-two law restricted to couplings (Thm. 10.13)	ill-typed; replaced by proposition 5.2(v)
Facets are cycles of $K_{n,n}$; 15 facets; smooth Fano fourfold	facets are cells; 9 facets; uniform 3×3 is not simple; nondegenerate square $n \geq 3$ is not Fano (proposition 6.3)
Lattice distances are sums over cycles	they are the entries of the coupling at the fibre position
Critical points of W are vertices (committed states)	critical points are interior torus fibres with flat connections (proposition 6.1); vertices are fixed points and optimal-transport optima (proposition 6.4)
$F_2 = \text{Hess } W$; Picard–Lefschetz formula; “symbolic proof”	withdrawn (section 8); the cited verification compared 0.1152 with 2
Cycle counts $N_{2k} = \frac{1}{2k} \binom{n}{k}^2 k!(k-1)!$; $K_{4,4}$ total 156	$N_{2k} = \binom{n}{k}^2 k!(k-1)!/2$; $K_{4,4}$ has $36 + 96 + 72 = 204$; $K_{5,5}$ has $100 + 600 + 1800 + 1440$
Bridgeland 2008 for all surfaces; “(i)–(iii) none known”	Bridgeland 2008 is for K3; general surfaces are [1] ; (i) is answered by proposition 7.1

B Computational checks

`check_cumulants.py` runs 8 checks for [section 9](#), all passing: Q1–Q3 for [propositions 9.1](#) and [9.2](#), and E1–E5 for [proposition 9.3](#).

`check_polytope_companion.py` runs 26 checks, all passing:

- chambers, types, surfaces, the sum rule, orbits, the del Pezzo fan, wall-crossings and [example 3.4](#) (F1–F7);
- [propositions 2.2](#) and [3.5](#) in all 18 chambers (K1–K3);
- [propositions 4.1](#) and [4.3](#) (S1–S4);
- [propositions 5.1](#) and [5.2](#) (L1–L7);
- [proposition 6.1](#), the non-monotone examples and [proposition 6.3](#) (W1–W3);

- [proposition 6.4](#) by linear programming (T1);
- the cycle counts of $K_{3,3}$ and $K_{4,4}$ by enumeration (C1).

Rational arithmetic is exact. The critical points are solved symbolically, and the linear programs use floating point.

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